

Nontrivial Case-3 Spin–Boson Renormalization

An explicit literature answer to arXiv:2508.00803

Literature-status note by `agent_lane_06`

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Status. `literature_already_answered`. This is a literature identification, not a new mathematical result of the run.

Source question

Alvarez, Lill, Lonigro, and Valentín Martín study generalized spin–boson Hamiltonians and divide ultraviolet singularities into three cases. Case 3 is the supercritical regime, in which at least one form factor f satisfies $f/\omega \notin L^2$. In Section 1.2, PDF page 3, after describing their triviality result, they write:

“The question whether it is possible to obtain a nontrivial Case 3 renormalization of certain GSB models remains open.”

This is an existential question: one nontrivially renormalizable class in the supercritical regime gives an affirmative answer.

Supporting answer

Falconi, Hinrichs, and Valentín Martín develop a wave-function renormalization on a renormalized, generally non-Fock Hilbert space. Their Theorem 3.18, “Renormalization of the SB model” (PDF page 22), applies to a bounded normal spin interaction operator B and every distributional source $v \in \mathcal{T}'$. It constructs a renormalized Hamiltonian $H_{\text{ren}}(B, v)$ and proves:

- self-adjointness on the dressed domain, with a specified core;
- a uniform lower bound by $-\|A\|$; and
- generalized strong resolvent convergence of regularized Hamiltonians as $v_\alpha \rightarrow v$ (and generalized norm resolvent convergence in the stated Hilbert-space subcase).

Because v is arbitrary in $\mathcal{T}'$, the theorem includes sources with $v/\omega \notin \mathfrak{h}$, precisely the supercritical regime excluded from energy-only renormalization on the original Fock space. The construction is therefore a nontrivial Case-3 renormalization and answers the source question affirmatively.

Provenance and scope

The identification is explicit in the literature. The supporting paper cites arXiv:2508.00803 in its account of critical and supercritical spin–boson renormalization. It supersedes the withdrawn arXiv:2508.00805, whose title and abstract explicitly announced that its construction solved the triviality problem for unitarily renormalized supercritical spin–boson models.

The result requires B to be bounded and normal. Thus, it does not provide a general theory for arbitrary nonnormal interaction matrices or arbitrary multi-interaction models. Those restrictions do not affect the affirmative answer to the source’s existential question.

Search evidence

The source bibliography already points to arXiv:2508.00805. Inspection of its current arXiv record showed that it was withdrawn in March 2026 and superseded by arXiv:2603.07045. The replacement paper’s abstract, Section 3 overview, Theorem 3.18, and citation to arXiv:2508.00803 establish the answer and its scope.

References

- [1] B. Alvarez, S. Lill, D. Lonigro, and J. Valentín Martín, *Renormalization of generalized spin–boson models with critical ultraviolet divergences*, arXiv:2508.00803v2 (2025), <https://arxiv.org/abs/2508.00803>.
- [2] M. Falconi, B. Hinrichs, and J. Valentín Martín, *Wave Function Renormalization for Particle-Field Interactions*, arXiv:2603.07045v1 (2026), <https://arxiv.org/abs/2603.07045>.
- [3] M. Falconi, B. Hinrichs, and J. Valentín Martín, *Non-Trivial Renormalization of Spin-Boson Models with Supercritical Form Factors*, withdrawn arXiv:2508.00805 (2025–2026), superseded by [2].